



1.9

Rational Functions and Vertical Asymptotes

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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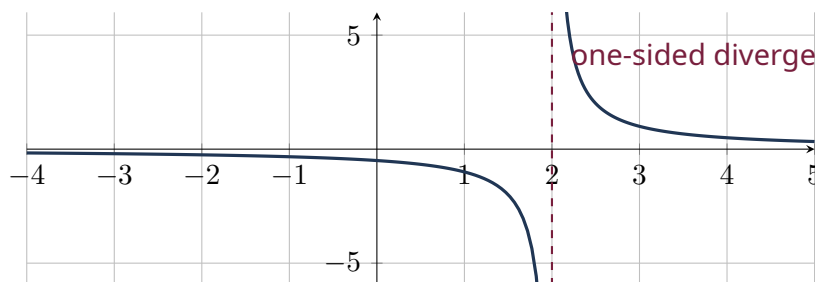
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Learning Targets

- Identify vertical asymptotes after simplifying a rational function.
- Describe one-sided behavior near vertical asymptotes.
- Use factor multiplicity and sign analysis to determine divergence.

Essential Question

How can local factor signs predict the direction of divergence near a vertical asymptote?



Concept Framework

Key Idea

A noncanceled denominator zero is a vertical-asymptote candidate.

Key Idea

Odd multiplicity changes the denominator sign across the asymptote; even multiplicity does not.

Key Idea

The numerator sign near the asymptote also matters.

Key Idea

One-sided statements are stronger and more precise than saying only “the function goes to infinity.”

Formula Map

Formula and Structure

$$\lim_{x \rightarrow a^-} R(x), \quad \lim_{x \rightarrow a^+} R(x)$$

Formula and Structure

$$Q(a) = 0, P(a) \neq 0 \Rightarrow x = a \text{ is a vertical asymptote}$$

Worked Examples

Worked Example 1

Problem. Describe one-sided behavior of $\frac{1}{x-2}$ at $x = 2$.

Solution. As $x \rightarrow 2^-$, denominator is small negative, so the function tends to $-\infty$. As $x \rightarrow 2^+$, it tends to $+\infty$.

Worked Example 2

Problem. Describe one-sided behavior of $-\frac{3}{(x+1)^2}$ at $x = -1$.

Solution. The squared denominator is positive on both sides and approaches 0; with negative numerator, both one-sided limits are $-\infty$.

Worked Example 3

Problem. Why must common factors be canceled before classifying denominator zeros?

Solution. A canceled factor produces a removable hole, not vertical divergence.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Find every vertical asymptote and state both one-sided behaviors for $\frac{2}{x-5}$.

2. Find every vertical asymptote and state both one-sided behaviors for $\frac{-3}{(x+2)^2}$.

3. Find every vertical asymptote and state both one-sided behaviors for $\frac{x+1}{(x-3)^3}$.

4. Find every vertical asymptote and state both one-sided behaviors for $\frac{x-4}{(x+1)^2}$.

5. Find every vertical asymptote and state both one-sided behaviors for $\frac{(x-2)^2}{x+4}$.

6. Analyze $\frac{x-1}{(x+2)(x-3)}$ near each vertical asymptote.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.